# Assignment 3
## 30 Marks
## Deadline: Jul 24, 2025 (Thursday)

1. An interpolating polynomial, **$p_1(x) = 1.648(x - 1)$** is derived for the function **$f(x) = x \ln x$** at the nodes **($x_0 = 1, x_1 = 5$)** using the Lagrange method. Answer the following keeping up to 4 significant figures.
   (a) (1 mark) Explain what you need to do to obtain a **degree 3** interpolating polynomial for the same function **f(x)** and for the same nodal points **($x_0 = 1, x_1 = 5$)**.
   (b) (4 marks) Calculate the bases of the **degree 3** polynomial. (Keep up to 4 significant figures for larger values)

2. Consider the following **Runge Function $f(x) = 1/(1+25x^2)$,** given interval **[-4, 6 ] and n = 2.** Answer the following using the given data,
   (a) [5 marks] Calculate the corresponding **equal angles**.
   (b) [5 marks] Calculate the **Chebyshev Nodes** up to 5 significant figures.

3. (3 + 2 Marks) The following Dataset is generated by the function **$f(x) = x\cos(x) - x + \sin(x)$**.

| | | | |
|---|---|---|---|
| x | 1.1 | 1.2 | 1.3 |
| f(x) | 0.2902 | 0.1669 | 0.01131 |

Based on the above data, compute f′(1.2) using the **Central Difference and Backward Difference** method, and also calculate the **relative error**. Use 4 significant figures. Also compute the upper bound of the truncation error of f(x) for **Backward Difference.**

4. Consider the function **$f(x) = x \ln(x)$**. Now answer the following:

(a) **[Ungraded]** Evaluate the numerical derivative of f(x) at x = 1.0 with step size h = 0.1 using the **forward and central difference** methods up to 5 significant figures.

(b) **[Ungraded]** Compute the upper bound of the truncation error of f(x) at x = 1.0 using h = 0.1 for the **backward and central difference** methods up to 5 significant figures.

(c) [10 Marks] Deduce an expression for $D_h^{(1)}$ by replacing h with **(4h/3)** using the Richardson Extrapolation method.